

## Paper 2 - Set 05 - Solutions

Generated practice paper: answer key and worked solutions.

### Answer key

| Question | Answer | Topic                                    |
|----------|--------|------------------------------------------|
| 1        | C      | Triangles, radians and sectors           |
| 2        | B      | Circles and circle theorems              |
| 3        | C      | Quadratic functions                      |
| 4        | A      | Chains of reasoning                      |
| 5        | A      | Chains of reasoning                      |
| 6        | C      | The logic of arguments                   |
| 7        | E      | The logic of arguments                   |
| 8        | E      | Mathematical proof                       |
| 9        | E      | Trigonometric functions and equations    |
| 10       | C      | Simultaneous equations and inequalities  |
| 11       | F      | Angles, polygons, congruence and vectors |
| 12       | C      | Identifying errors in proofs             |
| 13       | F      | Integration                              |
| 14       | H      | Indices and surds                        |
| 15       | A      | Mathematical proof                       |
| 16       | C      | Mathematical proof                       |
| 17       | E      | Chains of reasoning                      |
| 18       | A      | Identifying errors in proofs             |
| 19       | H      | Sequences and series                     |
| 20       | D      | Integration                              |

## Worked solutions

### Question 1 TRI-05 · Triangles, radians and sectors · spec MM4.2

Convert  $135^\circ$  to radians.

**Answer: C**

$$135 = \frac{3}{4} \times 180, \text{ so } \frac{3\pi}{4}.$$

#### Full method - how and why

Degree-to-radian conversion: multiply by  $\frac{\pi}{180}$ , or spot the fraction of  $180^\circ$ .

1 Express 135 as a fraction of 180:  $\frac{135}{180} = \frac{3}{4}$ .

why: Since  $180^\circ = \pi$  radians, the same fraction of  $\pi$  gives the answer.

2 So  $135^\circ = \frac{3}{4}\pi = \frac{3\pi}{4}$ .

why: Equivalently  $135 \times \frac{\pi}{180} = \frac{3\pi}{4}$ .

#### Where it goes wrong

- Multiplying by  $\frac{180}{\pi}$  (the reverse conversion).
- Simplifying  $\frac{135}{180}$  incorrectly to  $\frac{2}{3}$  or  $\frac{5}{6}$ , both of which are offered.

**Takeaway**  $180^\circ = \pi$ : convert by multiplying by  $\frac{\pi}{180}$ , and cancel the fraction fully.

### Question 2 CIR-10 · Circles and circle theorems · spec MM3.3

$TA$  is a tangent to a circle at  $A$ , and  $AB$  is a chord.  $C$  is a point on the circle on the opposite side of  $AB$  from  $T$ , with angle  $ACB = 65^\circ$ . Find angle  $TAB$ .

**Answer: B**

Alternate segment theorem: the angle between tangent and chord equals the angle in the alternate segment, so  $TAB = 65^\circ$ .

#### Full method - how and why

The alternate segment theorem, stated almost word for word. The work is in identifying which angle is 'in the alternate segment'.

1  $TA$  is tangent at  $A$  and  $AB$  is a chord, so the angle  $TAB$  between them is the tangent-chord angle.

why: The theorem applies exactly to this configuration.

2 The alternate segment is the one on the other side of  $AB$  from  $T$  - which is where  $C$  lies.

why: The question places  $C$  deliberately; the angle used must be in the segment **away** from the tangent's side.

3 Therefore angle  $TAB = \text{angle } ACB = 65^\circ$ .

why: Alternate segment theorem: the tangent-chord angle equals the inscribed angle in the alternate segment.

#### Where it goes wrong

- Using  $180^\circ - 65^\circ = 115^\circ$ , which is the angle on the other side of the tangent.
- Halving to  $32.5^\circ$  by confusing this with the centre-circumference theorem.

**Takeaway** Tangent-chord angle = angle in the alternate segment - the segment on the far side of the chord.

### Question 3 QUA-24 · Quadratic functions · spec MM1.3

Let  $f(x) = x^2 - 2x - 8$ . For how many integers  $n$  is  $f(n) < 0$ ?

**Answer: C**

$(x - 4)(x + 2) < 0$  gives  $-2 < x < 4$ ; the integers are  $-1, 0, 1, 2, 3$ : five of them.

#### Full method - how and why

Counting **integers** in the region where a quadratic is negative. Solve the inequality, then count carefully at the ends.

- 1 Factorise:  $n^2 - 2n - 8 = (n - 4)(n + 2)$ .

why: Two numbers multiplying to  $-8$  and adding to  $-2$ :  $-4$  and  $+2$ .

- 2 Negative between the roots:  $-2 < n < 4$ .

why: Upward parabola, so  $f < 0$  strictly between the roots.

- 3 List the integers strictly inside:  $-1, 0, 1, 2, 3$  - five of them.

why: The endpoints  $-2$  and  $4$  give  $f = 0$ , which is **not** less than zero, so they are excluded; and  $0$  is easy to forget.

#### Where it goes wrong

- Including  $-2$  and  $4$  and answering  $7$ .
- Forgetting  $0$  is an integer and answering  $4$ .

**Takeaway** After solving an inequality, list the integers explicitly - endpoints and zero are the two things people drop.

### Question 4 RSN-10 · Chains of reasoning · spec Prf5

Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?

**Answer: A**

The 50 pairs  $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$  each sum to 101. Fifty cards (one from each pair) may avoid it; 51 cannot.

#### Full method - how and why

Pigeonhole again, with the pairs built to sum to the target.

- 1 Pair the numbers to sum to 101:  $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$ .

why: Each of 1 to 100 lies in exactly one such pair, so there are 50 pairs.

- 2 Drawing one card from each pair gives 50 cards with **no** pair summing to 101 - for example 1 to 50.

why: So 50 is not enough; this constructs the worst case.

- 3 Any 51 cards must include two from one pair, and those sum to 101.

why: 51 cards into 50 pairs forces a repeat.

#### Where it goes wrong

- Answering 50 - the largest number that can fail.
- Answering 2 by thinking about luck rather than certainty.

**Takeaway** Same recipe as any pigeonhole guarantee: build the pairs, show the worst case, then add one.

### Question 5 RSN-05 · Chains of reasoning · spec Prf5

In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?

**Answer: A**

Each match eliminates one player and 36 must be eliminated, so 36 matches - whatever the draw looks like.

#### Full method - how and why

An invariant argument: count what each match achieves rather than simulating the draw.

- 1 At the end one player remains, so  $37 - 1 = 36$  players have been eliminated.  
*why:* Count the outcome, not the structure.
- 2 Each match eliminates exactly **one** player.  
*why:* Given in the question - this is the invariant.
- 3 So the number of matches equals the number of eliminations: 36.  
*why:* And this holds whatever byes or bracket shape are used, which is why the answer is unique.

#### Where it goes wrong

- Trying to build the bracket and getting lost in byes because 37 is not a power of 2.
- Halving repeatedly (18, 19-type answers), which only works for tidy brackets.

**Takeaway** Count the invariant: a knockout with  $n$  players always takes  $n - 1$  matches.

### Question 6 LGC-11 · The logic of arguments · spec Arg1

The statement '**if a triangle is equilateral then it is isosceles**' is true. Which of the following must therefore also be true?

- I If a triangle is isosceles then it is equilateral.
- II If a triangle is not isosceles then it is not equilateral.
- III If a triangle is not equilateral then it is not isosceles.

**Answer: C**

Only the contrapositive (II) is logically equivalent to the original. I is the converse and III is the inverse; both are in fact false here.

#### Full method - how and why

Given a true implication, only one of the three relatives is guaranteed. Identify which of I, II, III is the contrapositive.

- 1 I is 'isosceles  $\Rightarrow$  equilateral' - the **converse**. Not guaranteed, and here false (a 3, 3, 5 triangle).  
*why:* Swapped but not negated.
- 2 II is 'not isosceles  $\Rightarrow$  not equilateral' - the **contrapositive**. Always guaranteed. **True**.  
*why:* Swapped and negated; logically equivalent to the original.
- 3 III is 'not equilateral  $\Rightarrow$  not isosceles' - the **inverse**. Not guaranteed, and here false (the same 3, 3, 5 triangle).  
*why:* Negated but not swapped - and the inverse is just the contrapositive of the converse, so it carries the converse's truth value, not the original's.
- 4 So II only.  
*why:* Match to the list.

#### Where it goes wrong

- Accepting III because both its parts are negated - negation alone is not enough; the parts must also swap.
- Accepting I because equilateral triangles are also isosceles - that supports the original, not the converse.

**Takeaway** From  $A \Rightarrow B$  you get exactly one free statement:  $\neg B \Rightarrow \neg A$ .

### Question 7 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of ' $x > 3$  and  $y \leq 5$ '?

**Answer: E**

Not ( $A$  and  $B$ ) is (not  $A$ ) or (not  $B$ ). Not  $x > 3$  is  $x \leq 3$ ; not  $y \leq 5$  is  $y > 5$ .

#### Full method - how and why

De Morgan's law: negating an 'and' turns it into an 'or', with both parts negated.

1  $\neg(A \text{ and } B) \equiv (\neg A) \text{ or } (\neg B)$ .

why: For the conjunction to fail, it is enough that **one** part fails.

2 Negate the first part:  $\neg(x > 3)$  is  $x \leq 3$ .

why: The boundary  $x = 3$  satisfies the negation, since  $3 > 3$  is false.

3 Negate the second:  $\neg(y \leq 5)$  is  $y > 5$ .

why: Strict inequality, because  $y = 5$  satisfied the original.

4 Combine with 'or':  $x \leq 3$  **or**  $y > 5$ .

why: The joining word flips too - this is the half most often forgotten.

#### Where it goes wrong

- Keeping 'and': ' $x \leq 3$  and  $y > 5$ ' is far too strong - it demands both failures.
- Getting the boundaries wrong: writing  $x < 3$  or  $y \geq 5$ , which mishandles  $x = 3$  and  $y = 5$ .

**Takeaway** De Morgan:  $\text{not}(A \text{ and } B) = \text{not}A \text{ or } \text{not}B$ ;  $\text{not}(A \text{ or } B) = \text{not}A \text{ and } \text{not}B$ . Negate the connective as well as the parts.

### Question 8 PRF-01 · Mathematical proof · spec Prf1

Consider the claim: **for every positive integer  $n$ ,  $n^2 + n$  is even**. Which of the following is a complete and valid proof?

**Answer: E**

The factorised argument covers every  $n$  at once:  $n(n + 1)$  is a product of consecutive integers, so one factor is even. Checking  $n = 1, 2, 3$  is only three cases; 'if  $n$  is even ...' leaves odd  $n$  untreated; 'both  $n^2$  and  $n$  are even' is false for odd  $n$ ; and the contradiction attempt ends with an unresolved exception.

#### Full method - how and why

What makes a proof **complete**: it must cover every case in the claim with a general argument, not a sample of values.

1 The claim is about **every** positive integer, so any valid proof must handle all  $n$  at once.

why: Checking particular values can only ever disprove a universal claim, never prove one.

2 Factorise:  $n^2 + n = n(n + 1)$  - the product of two **consecutive** integers.

why: Factorising turns a statement about a sum into one about a product, where parity is easy to control.

3 Of any two consecutive integers, exactly one is even; an even factor makes the product even.

why: This single sentence covers both the case  $n$  even and the case  $n$  odd - no case-splitting needed.

4 That argument is general and complete, so it is the proof.

why: Now examine why the others fail, which is the real content of the question.

#### Where it goes wrong

- Accepting the check of  $n = 1, 2, 3$ : three cases out of infinitely many prove nothing.
- Accepting 'if  $n$  is even then  $n^2 + n$  is even' - true, but it leaves odd  $n$  untreated, so the proof is incomplete.
- Accepting 'both  $n^2$  and  $n$  are even' - simply false when  $n$  is odd.
- Accepting the contradiction attempt that ends 'unless  $n$  is even' - a proof cannot end with an unresolved exception.

**Takeaway** A proof of 'for all  $n$ ' must be general. Examples illustrate; only argument proves.

**Question 9** X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation  $\sin x = k$  where  $k$  is a real constant. Which of the following statements is/are true?

- I For every  $k$  with  $-1 < k < 1$  the equation has exactly two solutions in  $0 \leq x < 2\pi$ .
- II For  $k = 1$  the equation has exactly one solution in  $0 \leq x < 2\pi$ .
- III For every real  $k$  the equation has at least one solution.

**Answer: E**

I and II are read from the graph of  $\sin$  over one period. III is false:  $\sin x = 2$  has no solutions.

**Full method - how and why**

Read the claims off the graph of  $y = \sin x$  over one period, then remember the range.

- 1 I: for  $-1 < k < 1$  the horizontal line  $y = k$  cuts one period of the sine curve exactly twice. **True.**  
why: Sine rises from 0 to 1, falls to  $-1$  and returns - a strictly-between value is hit once on the way down and once elsewhere in  $[0, 2\pi)$ .
- 2 II:  $k = 1$  is reached only at the single peak  $x = \frac{\pi}{2}$ . **True.**  
why: The maximum is attained once per period.
- 3 III: for  $|k| > 1$  there is no solution at all -  $\sin x = 2$  is impossible. **False.**  
why: The range of sine is  $[-1, 1]$ .
- 4 So I and II.  
why: Match to the list.

**Where it goes wrong**

- Accepting III by forgetting the range restriction.
- Rejecting II by counting  $x = \frac{\pi}{2}$  and  $x = \frac{5\pi}{2}$  - the second is outside  $0 \leq x < 2\pi$ .
- Rejecting I because  $k = 0$  feels special:  $\sin x = 0$  gives  $x = 0$  and  $x = \pi$  in the interval - still exactly two.

**Takeaway** Per period,  $\sin x = k$  has two solutions for  $|k| < 1$ , one for  $|k| = 1$  and none for  $|k| > 1$ . Sketch the line across the curve.

**Question 10** X2-14 · Simultaneous equations and inequalities · spec MM1.5

$x$  is a real number.  $P$ : ' $x^2 < 4$ '.  $Q$ : ' $x < 2$ '. Which one of the following is true?

**Answer: C**

$x^2 < 4$  means  $-2 < x < 2$ , which implies  $x < 2$  (sufficient); but  $x = -5$  satisfies  $Q$  and not  $P$  (not necessary).

**Full method - how and why**

Solve the quadratic inequality properly, then compare the two solution sets as intervals.

- 1  $x^2 < 4$  means  $|x| < 2$ , i.e.  $-2 < x < 2$ .  
why: A quadratic inequality of this shape gives an interval **between** the roots.
- 2 Every  $x$  in  $(-2, 2)$  satisfies  $x < 2$ , so  $P \Rightarrow Q$ : **sufficient**.  
why:  $(-2, 2)$  is contained in  $(-\infty, 2)$ .
- 3 Is it necessary? Take  $x = -5$ :  $x < 2$  holds but  $x^2 = 25 < 4$ . So  $Q \Rightarrow P$ : **not necessary**.  
why: The larger set contains points outside the smaller one.
- 4 So  $P$  is sufficient but not necessary.  
why: Whenever one solution set sits strictly inside another, the inner condition is sufficient but not necessary.

**Where it goes wrong**

- Solving  $x^2 < 4$  as  $x < 2$  and concluding the conditions are equivalent - that drops the lower bound.
- Solving it as  $x < -2$  or  $x > 2$ , which is the solution of  $x^2 > 4$ .

**Takeaway** Compare conditions as **sets**: a subset is sufficient for the superset, the superset necessary for the subset.

### Question 11 GEO-10 · Angles, polygons, congruence and vectors · spec M5.3

Which of the following statements is/are true?

- I Every rhombus is a parallelogram.
- II Every rectangle is a rhombus.
- III The diagonals of a kite are perpendicular.

**Answer: F**

A rhombus has two pairs of parallel sides (I). A rectangle need not have equal sides (II false). The diagonals of a kite cross at right angles (III).

#### Full method - how and why

Quadrilateral classification. Each claim is about whether one family is contained in another.

- 1 I: a rhombus has two pairs of parallel sides, which is the definition of a parallelogram. **True.**  
why: Rhombus = parallelogram with all sides equal, so it is a special parallelogram.
- 2 II: a rectangle has equal **angles** but not necessarily equal sides - a  $2 \times 5$  rectangle is not a rhombus. **False.**  
why: Only the square lies in both families.
- 3 III: a kite's diagonals meet at right angles (the axis of symmetry bisects the other diagonal perpendicularly). **True.**  
why: This is a standard kite property.
- 4 I and III only.  
why: Match to the option list.

#### Where it goes wrong

- Reversing II: every rhombus is not a rectangle either, but the claim as stated is about rectangles.
- Thinking a kite's diagonals **bisect** each other - only one bisects the other.

**Takeaway** Square = rectangle  $\cap$  rhombus; kites have perpendicular diagonals but only one is bisected.

### Question 12 ERR-14 · Identifying errors in proofs · spec Err2

Claim: if  $x$  and  $y$  are real with  $x^2 = y^2$ , then  $x = y$ .

Which of the following is a counterexample to the claim?

**Answer: C**

A counterexample must satisfy the hypothesis ( $x^2 = y^2$ ) and fail the conclusion ( $x \neq y$ ).

#### Full method - how and why

A counterexample must satisfy the hypothesis and break the conclusion. Test both halves on each candidate.

- 1 Hypothesis:  $x^2 = y^2$ . Conclusion to break: need  $x \neq y$ .  
why: Both conditions, or it is not a counterexample.
- 2  $x = 2, y = -2$ :  $4 = 4$  ✓ hypothesis, and  $2 \neq -2$  ✓ conclusion fails. **Counterexample.**  
why: Equal squares with opposite signs is exactly the gap in the claim.
- 3  $x = 2, y = 2$  and  $x = 0, y = 0$  and  $x = -1, y = -1$ : hypothesis holds but so does the conclusion - these are examples of the claim.  
why: Confirming instances never refute.
- 4  $x = 1, y = 2$ :  $1 \neq 4$ , so the hypothesis fails and the case is irrelevant.  
why: A case outside the hypothesis says nothing.

#### Where it goes wrong

- Picking a pair where the hypothesis fails and thinking the different values refute the claim.
- Picking a pair where both hold, which supports rather than refutes.

**Takeaway**  $x^2 = y^2 \Rightarrow x = \pm y$ . A counterexample satisfies the 'if' and fails the 'then'.

**Question 13** INT-09 · Integration · spec MM7.1

Which of the following statements is/are true?

**I**  $\int_{-2}^2 x^3 dx = 0$

**II** The area between  $y = x^3$  and the  $x$ -axis from  $x = -2$  to  $x = 2$  is 0.

**III**  $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

**Answer: F**

I: odd function over a symmetric interval, so 0. II is false: the area is  $2 \int_0^2 x^3 dx = 8$ ; the integral is 0 because the parts cancel. III: even function, true.

**Full method - how and why**

Three statements distinguishing **signed integrals** from **areas**, using odd and even symmetry.

**1** I:  $x^3$  is odd, so the contributions from  $[-2, 0]$  and  $[0, 2]$  cancel exactly: the integral is 0. **True.**

*why:* For odd  $f$ ,  $\int_{-a}^a f = 0$  always.

**2** II: the **area** counts both pieces positively:  $2 \int_0^2 x^3 dx = 2 \times 4 = 8 \neq 0$ . **False.**

*why:* Area and integral differ precisely when the curve crosses the axis - this is the distinction the question is testing.

**3** III:  $x^2$  is even, so the two halves are equal:  $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$ . **True.**

*why:* For even  $f$ ,  $\int_{-a}^a f = 2 \int_0^a f$ .

**4** I and III only.

*why:* Match to the option list.

**Where it goes wrong**

- Treating 'integral' and 'area' as synonyms and marking II true.
- Thinking odd symmetry also halves the area.

**Takeaway** Odd function over a symmetric interval: integral 0 but area positive. Even function: integral is twice the half.

**Question 14** X2-12 · Indices and surds · spec MM1.1

$a$  and  $b$  are positive real numbers. Which of the following statements is/are true?

- I If  $a^2 = b^2$  then  $a = b$ .
- II If  $2^a = 2^b$  then  $a = b$ .
- III If  $a^b = 1$  then  $a = 1$ .

**Answer: H**

I holds because  $a, b > 0$  rules out  $a = -b$ . II holds because  $2^x$  is one-to-one. III: for positive  $a$  and  $b$ ,  $a^b = 1$  gives  $b \log a = 0$ , so  $\log a = 0$  and  $a = 1$ . All three are true.

**Full method - how and why**

Three cancellation claims, all made safe by the stated positivity. Check what each would need.

- 1 I:  $a^2 = b^2$  gives  $a = \pm b$  in general - but  $a, b > 0$  rules out  $a = -b$ , so  $a = b$ . **True.**  
why: The positivity hypothesis is exactly what repairs the usual counterexample.
- 2 II:  $2^x$  is a strictly increasing function, so it takes each value once:  $2^a = 2^b \Rightarrow a = b$ . **True.**  
why: One-to-one functions can be cancelled; equivalently take  $\log_2$  of both sides.
- 3 III:  $a^b = 1$  with  $a, b > 0$ . Take logs:  $b \log a = 0$ . Since  $b > 0$ ,  $\log a = 0$ , so  $a = 1$ . **True.**  
why: Taking logs turns the exponent into a factor, and a product is zero only if a factor is.
- 4 So all three.  
why: Match to the list.

**Where it goes wrong**

- Rejecting I by recalling the counterexample 2 and  $-2$  - forbidden here, since both are positive.
- Rejecting III by thinking of  $a^0 = 1$  - that needs  $b = 0$ , which positivity forbids. (And  $(-1)^2 = 1$  needs a negative  $a$ .)
- Overlooking how much work the phrase 'positive real numbers' is doing in all three.

**Takeaway** Read the domain restrictions first - positivity makes squaring and exponentiation reversible, when in general they are not.

**Question 15** PRF-11 · Mathematical proof · spec Prf3

A student notices that  $1 + 2 = 3$ ,  $4 + 5 + 6 = 7 + 8$ ,  $9 + 10 + 11 + 12 = 13 + 14 + 15$  and conjectures that the pattern continues. Which of the following describes the  $n$ th line of the pattern?

**Answer: A**

Line 1 starts at 1, line 2 at 4, line 3 at 9: start  $n^2$ . Left sides have 2, 3, 4 numbers ( $n + 1$ ), right sides 1, 2, 3 ( $n$ ).

**Full method - how and why**

Describing a pattern precisely: check the start value and the two counts separately against every given line.

- 1 Starting values: 1, 4, 9 for lines 1, 2, 3 - these are  $n^2$ .  
why: Test against all three lines, not just one.
- 2 Count the left sides:  $\{1, 2\}$ ,  $\{4, 5, 6\}$ ,  $\{9, 10, 11, 12\}$  - that is 2, 3, 4 terms, i.e.  $n + 1$ .  
why: Count carefully; being off by one here is the whole point of the question.
- 3 Count the right sides:  $\{3\}$ ,  $\{7, 8\}$ ,  $\{13, 14, 15\}$  - that is 1, 2, 3 terms, i.e.  $n$ .  
why: Left has one more term than right, which is what makes the equality possible (the right-hand terms are larger).
- 4 So: starts at  $n^2$ , with  $n + 1$  terms on the left and  $n$  on the right.  
why: Check the total count:  $(n + 1) + n = 2n + 1$ , and indeed line 3 runs from 9 to 15, seven numbers ✓.

**Where it goes wrong**

- Swapping the counts ( $n$  left,  $n + 1$  right) - the sides would not balance, since the right-hand numbers are bigger.
- Reading the start as  $2n - 1$  or  $n(n + 1)$  - both fail at line 2 (3 and 6 rather than 4).

**Takeaway** Verify a conjectured pattern against **every** given line, and check each component (start, length) separately.

### Question 16 PRF-06 · Mathematical proof · spec Prf1

Which of the following statements can be disproved by a single counterexample?

- I Every prime number greater than 2 is odd.
- II Every integer of the form  $4k + 3$  is prime.
- III There exists an even prime number.

**Answer: C**

Only a 'for all' statement can be disproved by one counterexample. I is true (no counterexample exists). II is false and  $k = 3$  (15) disproves it. III is an existence statement - a counterexample is meaningless; it is proved by the example 2.

#### Full method - how and why

Only a **universal** statement that is **false** can be disproved by a counterexample. Two separate tests.

- 1 I: 'every prime  $> 2$  is odd' is universal - but it is **true** (any even number  $> 2$  has factor 2), so no counterexample exists.  
*why:* You cannot disprove a true statement.
- 2 II: 'every integer  $4k + 3$  is prime' is universal and **false**:  $k = 3$  gives  $15 = 3 \times 5$ . Disprovable. **Yes**.  
*why:* One failing case is a counterexample.
- 3 III: 'there exists an even prime' is **existential**, not universal - and it is true (2). A counterexample is not even the right tool.  
*why:* To disprove an existential statement you would have to rule out every case, which is not a 'single counterexample'.
- 4 So II only.  
*why:* Match to the list.

#### Where it goes wrong

- Including I because it is a 'for all' statement - the statement also has to be false.
- Including III by confusing what disproves an existential claim.

**Takeaway** Counterexamples disprove false universal statements only. Existential statements are proved by example and disproved only by a general argument.

**Question 17** RSN-19 · Chains of reasoning · spec Prf5

A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

**Answer: E**

With  $r$  red and  $b$  blue:  $r - 1 = \frac{1}{4}(r + b - 1)$  and  $r = \frac{1}{3}(r + b - 2)$ . The second gives  $b = 2r + 2$ ; substituting into the first,  $4r - 4 = 3r + 1$ , so  $r = 5$ ,  $b = 12$ : 17 balls.

**Full method - how and why**

Two conditions, two unknowns: translate each sentence into an equation, then solve simultaneously.

- 1 Let  $r$  be red and  $b$  blue. Removing one red:  $r - 1 = \frac{1}{4}(r + b - 1)$ .

why: The total drops by 1 as well as the red count - both sides of the fraction change.

- 2 Removing two blues:  $r = \frac{1}{3}(r + b - 2)$ .

why: Here the red count is unchanged; only the total falls, by 2.

- 3 From the second:  $3r = r + b - 2$ , so  $b = 2r + 2$ .

why: Solve the simpler equation first and substitute.

- 4 Substitute into the first:  $4(r - 1) = r + (2r + 2) - 1 = 3r + 1$ , so  $4r - 4 = 3r + 1$  and  $r = 5$ .

why: Multiply through by 4 to clear the fraction before substituting.

- 5 Then  $b = 12$  and the total is 17.

why: Check: remove one red  $\rightarrow$  4 red of 16  $= \frac{1}{4} \checkmark$ ; remove two blue  $\rightarrow$  5 red of 15  $= \frac{1}{3} \checkmark$ .

**Where it goes wrong**

- Forgetting that removing a ball also reduces the **total**, giving  $r - 1 = \frac{1}{4}(r + b)$ .
- Answering  $r$  or  $b$  rather than the total.
- Not checking the answer against both original statements.

**Takeaway** When items are removed, adjust the total as well as the category; then verify against both conditions.

**Question 18** ERR-05 · Identifying errors in proofs · spec Err2

A student solves the inequality  $\frac{x+1}{x-2} < 3$ .

**I** Multiply both sides by  $(x-2)$ :  $x+1 < 3x-6$ .

**II** So  $7 < 2x$ .

**III** So  $x > \frac{7}{2}$ .

Which of the following is correct?

**Answer: A**

For  $x < 2$  the multiplier is negative and the inequality reverses; indeed  $x = 0$  gives  $-\frac{1}{2} < 3$ , a solution missed by the student. The full solution set is  $x < 2$  or  $x > \frac{7}{2}$ .

**Full method - how and why**

Multiplying an inequality by an expression whose sign is unknown. The error is the unstated assumption.

**1** Line I multiplies by  $(x-2)$  and keeps the direction of the inequality.

*why:* That is only valid if  $x-2 > 0$ ; if  $x < 2$  the inequality must reverse.

**2** So line I is the **first error**: it silently assumes  $x > 2$ .

*why:* The line is not wrong for all  $x$ , but it is wrong as an unconditional step - which is what makes it an error.

**3** Consequence: the student misses a whole branch. Test  $x = 0$ :  $\frac{0+1}{0-2} = -\frac{1}{2} < 3$  ✓ - a genuine solution not in  $x > \frac{7}{2}$ .

*why:* A single test value exposes the omission.

**4** The correct solution set is  $x < 2$  or  $x > \frac{7}{2}$ .

*why:* For  $2 < x < \frac{7}{2}$  the fraction exceeds 3; at  $x = 2$  it is undefined.

**Where it goes wrong**

- Saying you can never multiply an inequality by an expression - you can, if you case-split on its sign, or multiply by  $(x-2)^2$ , which is always positive.
- Blaming line II or III; the arithmetic there is right, given line I.
- Forgetting that  $x = 2$  must be excluded from any solution set.

**Takeaway** Never multiply an inequality by an expression of unknown sign: split into cases, or multiply by its square.

**Question 19** SEQ-21 · Sequences and series · spec MM2.1

A sequence has  $u_1 = 3$  and  $u_{n+1} = \frac{u_n + 5}{2}$ . Which of the following statements is/are true?

**I** The sequence is increasing.

**II**  $u_n < 5$  for all  $n$ .

**III**  $u_n = 5 - 2^{2-n}$  for all  $n$ .

**Answer: H**

$u_2 = 4$ ,  $u_3 = 4.5$ ,  $u_4 = 4.75$ . If  $u_n < 5$  then  $u_{n+1} = \frac{u_n + 5}{2} < 5$  and  $u_{n+1} - u_n = \frac{5 - u_n}{2} > 0$ , so I and II hold. III:  $5 - 2^{2-n}$  gives 3, 4, 4.5, ... and satisfies the recurrence:  $\frac{5 - 2^{2-n} + 5}{2} = 5 - 2^{1-n}$ . All three are true.

**Full method - how and why**

A recurrence converging to a limit. Compute terms for intuition, then prove the two general claims and verify the closed form.

**1** Generate terms:  $u_1 = 3$ ,  $u_2 = \frac{3+5}{2} = 4$ ,  $u_3 = 4.5$ ,  $u_4 = 4.75$  - rising towards 5.

why: The rule averages  $u_n$  with 5, so each term sits halfway between the previous one and 5.

**2** II: show  $u_n < 5$  always. If  $u_n < 5$  then  $u_{n+1} = \frac{u_n + 5}{2} < \frac{5+5}{2} = 5$ ; and  $u_1 = 3 < 5$ . **True.**

why: This is an induction argument in miniature: true at the start, and preserved by the rule.

**3** I:  $u_{n+1} - u_n = \frac{u_n + 5}{2} - u_n = \frac{5 - u_n}{2} > 0$  because  $u_n < 5$ . So the sequence increases. **True.**

why: Compute the difference algebraically rather than relying on the first few terms; it also shows \*why\* it increases - the gap to 5 is always positive.

**4** III: test  $5 - 2^{2-n}$ : at  $n = 1$  it gives  $5 - 2 = 3$  ✓, and  $\frac{(5 - 2^{2-n}) + 5}{2} = 5 - 2^{1-n}$ , which is the formula at  $n + 1$  ✓. **True.**

why: Check the starting value **and** that the formula satisfies the recurrence - both are needed.

**5** All three are true.

why: Match to the options.

**Where it goes wrong**

- Concluding II from the first four terms alone - the general argument is what makes it certain.
- Checking III only at  $n = 1$ ; many formulas share a first term.

**Takeaway** To prove a claim about every term: check the first case, then show the recurrence preserves the property.

**Question 20** X2-03 · Integration · spec MM7.1

$f$  is a function defined for all real  $x$  and  $\int_0^4 f(x) dx = 0$ . Which of the following statements must be true?

- I  $f(x) = 0$  for some  $x$  with  $0 \leq x \leq 4$ .
- II The area between the curve and the  $x$ -axis from 0 to 4 is 0.
- III  $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

**Answer: D**

III is just the rule for combining integrals over adjacent ranges. I fails for a function that jumps from  $-1$  to  $1$  without passing through 0 (nothing says  $f$  is continuous). II fails:  $f(x) = x - 2$  has integral 0 but area 4.

**Full method - how and why**

A hard 'must be true' about integrals. Two claims import assumptions the question never makes.

- 1 III:  $\int_0^4 = \int_0^2 + \int_2^4$ , and the left side is 0, so  $\int_0^2 = -\int_2^4$ . **True** - pure algebra of integrals.

why: Splitting an integral at an interior point is always valid where the integral exists.

- 2 I: 'zero integral  $\Rightarrow f$  hits zero' is the intermediate value theorem in disguise - but that needs **continuity**, which is not given.

why: Read the hypotheses: it only says  $f$  is defined for all real  $x$ .

- 3 Counterexample for I: a step function equal to  $-1$  on  $[0, 2)$  and  $1$  on  $[2, 4]$  has integral 0 but never equals 0.

**False.**

why: It jumps straight past zero.

- 4 II: 'integral 0' means signed areas cancel, not that the area is 0. Take  $f(x) = x - 2$ :  $\int_0^4 (x - 2) dx = 0$ , but the area is  $2 + 2 = 4$ . **False.**

why: Area counts the region below the axis as positive; the integral counts it as negative.

- 5 So III only.

why: Match to the list.

**Where it goes wrong**

- Accepting I by silently assuming  $f$  is continuous - an assumption the question deliberately withholds.
- Treating 'integral' and 'area' as the same thing; they agree only when  $f \geq 0$  throughout.

**Takeaway** Integral  $\neq$  area when the function changes sign; and the intermediate value theorem requires continuity, which must be **given**.